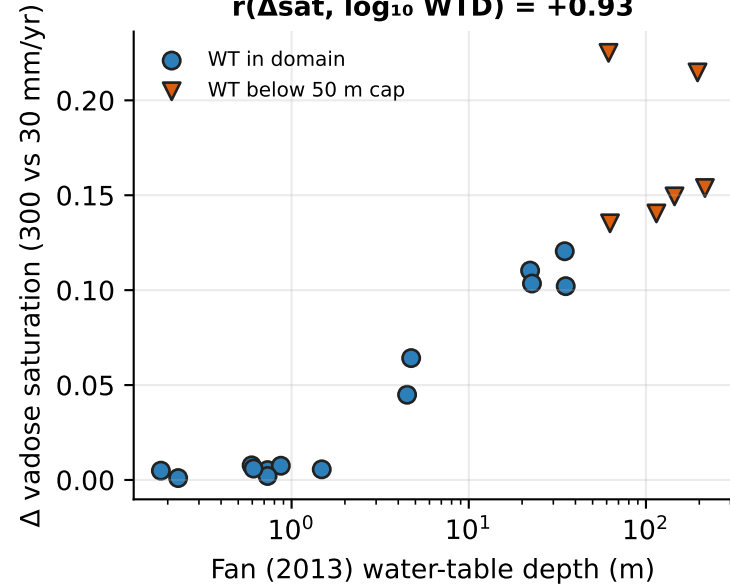
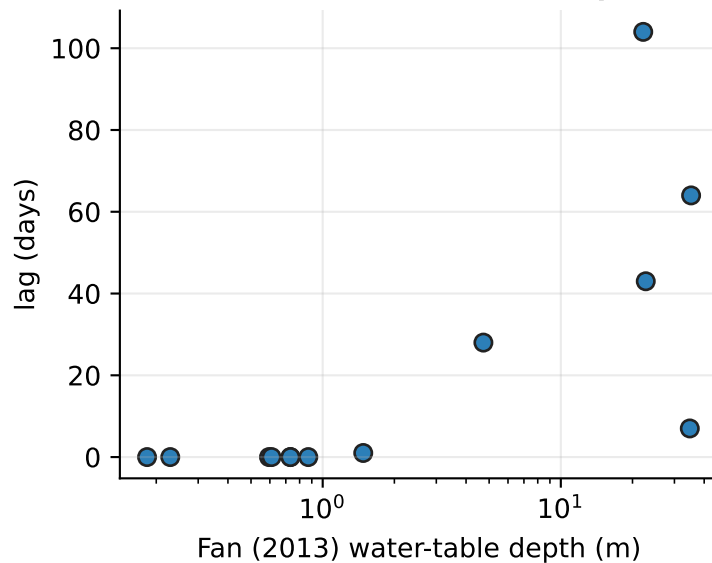


The vadose zone as a low-pass filter — cutoff set by water-table depth (neither model shows this alone)

(a) Standalone sweep — climate sensitivity
 $r(\Delta\text{sat}, \log_{10} \text{WTD}) = +0.93$



(b) Coupled — infiltration → water-table lag
13 columns with coherent response



(c) Coupled — signal surviving the vadose zone
 $r(\text{attenuation}, \log_{10} \text{WTD}) = -0.90$

